

* If you put garbage in, you get garbage out *

Data engineers stop the garbage from getting in

How do data engineers wash that "mud" away?

E-T-L

① E = Extract (The "Go-Get-It" step)

- Engineers pipes go out & scoop up data from everywhere - like how many people clicked a button on a website, or what time someone bought a toy.

② T = Transform (Washing & chopping step)

- Wash: data engineer washes off the real dirt (deletes blanks or broken information)
- Chop: Some carrots are giant & some tiny; they are chopped to be the same size. If someone wrote their name as "SAM", "sam" or "S-a-m", the engineer fixes it so it looks neat: "Sam".
- Mix: Measure everything perfectly so it's ready to become a soup.

L = Load ("Put-it-away step")

Veggies are now clean; chopped and perfect. They get poured into a giant warm soup pot (Data Warehouse)

Pot keep the soup perfectly safe & ready.

When the hungry chef (Data scientist)

comes into the kitchen, they don't have to wash dirt from a single carrot. They just take a ladle, scoop out the perfect soup & taste it.

Extract (Get it); Transform (Clean it)

Load (Put it away)

OLTP

OLAP

ELT

Data engineer

DATA WAREHOUSE

DB1

DB2

DB3